

Answers:

Question 1:

For the LogNormal distribution, based on my monte carlo, the expected annual damage is \$115,004, meaning this is the average yearly loss we would anticipate over the long term. The 99% quantile damage is \$189,761, representing the damage level that would be exceeded in only 1% of years.

For the Generalized Extreme Value (GEV) distribution, based on my monte carlo, the expected annual damage is slightly lower at \$113,360, while the 99% quantile damage is \$183,314. This indicates that the GEV distribution predicts somewhat lower risk overall, particularly for extreme events.

Question2:

I tested various sample sizes (1,000; 10,000; 50,000; 100,000) and observed that the estimates stabilized around 50,000-100,000 samples. With 100,000 simulations, the standard error for the expected value is approximately \$108, providing sufficient precision for risk management decisions while maintaining reasonable computation time.

Question3:

I believe they are different with each other. Although two results show almost the same, as two distributions have centralized at a similar mean (about 3m), however, the two distributions, LogNormal and GEV, have fundamentally different tail behaviors and skewness characteristics. The LogNormal distribution exhibits right-skewness, meaning it has a heavier upper tail that assigns higher probability to extreme flood events.

Question4:

I believe that the choice should be based on our needs and can be changed accordingly. First, we need to consider the mechanism of the floods in that area. If the floods are caused by a multiplicative process (such as rainfall intensity $\times$ duration $\times$ basin characteristics), then the log-normal distribution might be more applicable. However, if we specifically model the annual maximum values, the gamma-exponential distribution has theoretical advantages.